

Automated Logical Deciders

From theorem proving to certified conjecture settlement

Concept proposed by **Brian Tenneson**; formalized in collaboration with ChatGPT | Draft 001 | August 1, 2026

An automated theorem prover asks whether a theory T proves a sentence c . An **Automated Logical Decider (ALD)** asks the larger question: **What is the certified logical status of c relative to T ?**

1. Core definition

An ALD receives an object theory T , a conjecture c , a named metatheory M , and a resource budget B . It searches for checkable certificates supporting one of the statuses below. The phrase *relative to T* is indispensable: a sentence can be independent of one theory and decided by a stronger theory.

STATUS	REQUIRED CERTIFICATE	MEANING
PROVED	A formal proof of c from T .	$T \vdash c$
REFUTED	A formal proof of not- c from T .	$T \vdash \text{not-}c$
INDEPENDENT	A metatheoretic proof, checked in M , that neither c nor not- c is provable from T ; equivalently, under stated assumptions, certified models or consistency proofs for both $T + c$ and $T + \text{not-}c$.	c is independent of T , as certified in M .
INCONSISTENT	A formal proof of contradiction from T .	$T \vdash \text{false}$
UNSETTLED	No successful certificate within budget B .	No mathematical conclusion.

2. The decisive rule

Failure to find a proof is never a proof of independence. Independence is returned only when the ALD produces an independently checkable metatheoretic certificate.

Cohen's lesson

The Continuum Hypothesis was not abandoned when proof and refutation within ZFC failed. Goedel established the relative consistency of CH with ZFC, and Cohen established the relative consistency of its negation. Together, subject to the usual consistency assumptions, they settled CH by showing that it is independent of ZFC. For an ALD, independence is therefore a successful settlement, not a sophisticated timeout.

3. Certificate-producing architecture

A practical ALD is an orchestrated population of search engines. Each branch may use different representations and heuristics, but every accepted result must be translated into a certificate for a small trusted verifier.

COMPONENT	TASK	TYPICAL ENGINE
Positive	Search for $T \mid -c$.	Predator 1
Negative	Search for $T \mid -\text{not-}c$.	Predator 2
Model +	Certify a model of $T + c$.	Predator 3 with model finder
Model -	Certify a model of $T + \text{not-}c$.	Predator 4 with model finder
Metatheory	Combine models, consistency, forcing, or interpretation results into independence.	Predator 5 working in M
Coordinator	Allocate budgets, route typed lemmas, and detect complete certificate combinations.	Small non-mathematical traffic director

Certificate discipline

- Discovery modules may behave as black boxes; accepted conclusions may not.
- Object-theory and metatheory lemmas must occupy separate, typed stores.
- Cross-theory transfers require verified interpretations or translations.
- Proof sharing is permitted only after verification and with provenance preserved.

4. What an ALD cannot promise

For sufficiently expressive, effectively presented theories, no universal procedure can halt on every sentence and always classify it correctly as proved, refuted, or independent. Undecidability and incompleteness rule out that machine. Accordingly, the general research target is a **certificate-producing partial ALD**: it may return UNSETTLED, but it never promotes search failure into a theorem. In restricted decidable logics, a specialized ALD may be total and complete.

5. Predator's role

Predator is the proof-search intelligence inside an ALD, not the entire ALD. In the full design, five specialist Predators pursue proof, refutation, two opposite model constructions, and the final metatheoretic independence proof. A lightweight coordinator directs traffic without asserting mathematical claims. It manages budgets, keeps theory levels distinct, and reports only statuses supported by certificates.

6. Research objective

Settle conjectures no human has settled - by proving them, refuting them, or proving their independence from explicitly named axioms.

The proposed progression is: **automated theorem prover** -> **automated conjecture settler** -> **Automated Logical Decider**. The first searches for a proof. The second searches symmetrically for proof or refutation. The third can also change logical levels and seek a certified explanation of why neither side follows from the chosen theory.

Terminological note: in computability theory, a decider is normally total. The name ALD expresses the intended task; unless a decidable domain is specified, the technically exact implementation is a partial ALD with explicit resource bounds and certificate requirements.